

MECHSOURCE PRODUCT REQUIREMENTS DOCUMENT

Product Name: MECHSOURCE

Tagline: Find the Right Part. Find the Right Supplier. Get it Delivered.

Document: 01 Product Requirements Document

Status: Product Definition

Purpose: Master product reference for research, UX/UI design, architecture and future development

1. PRODUCT VISION

MECHSOURCE is a digital infrastructure platform for vehicle, machinery and equipment maintenance.

The platform will connect customers with spare parts suppliers, equipment suppliers, mechanics, technicians, workshops, rental providers, logistics providers and industrial businesses through one connected digital ecosystem.

MECHSOURCE is not intended to be only an online spare parts store.

The long term vision is to create a platform where a customer can identify a required part, find compatible products, compare suppliers, purchase the part, arrange delivery, find a technician, maintain the vehicle or equipment and manage its maintenance history from one platform.

The platform should eventually support individuals, workshops, transport businesses, fleet operators, construction companies, mining companies, industrial organisations and procurement teams.

2. CORE PROMISE

MECHSOURCE exists to solve three major problems:

1. Difficulty identifying the correct part
2. Difficulty finding a reliable supplier
3. Difficulty obtaining the part quickly and conveniently

The core promise is:

- Find the right part.
- Find the right supplier.
- Get it delivered.

3. CORE PRODUCT IDEA

The central experience of MECHSOURCE is called:

FIND MY PART

A customer should be able to provide information about a vehicle or machine and identify the required part.

Example (just a sample):

Vehicle: Toyota Hilux

Year: 2018

Engine: 2.8L Diesel

Required Part: Fuel Filter

MECHSOURCE should then help the customer discover:

- Part name
- OEM number
- Manufacturer number
- Compatible alternatives
- Compatible vehicle or equipment
- Available suppliers
- Price
- Stock status
- Supplier location
- Delivery options
- Estimated delivery time
- Supplier rating
- Warranty or return information where applicable

The customer should also be able to upload a photograph of an unknown or damaged part.

The system can then use AI assisted image identification to suggest the likely part.

The AI result must be treated as an identification recommendation rather than an automatic guarantee.

Where confidence is low, the platform should allow additional information to be provided or allow the customer to request verification from a qualified supplier or technician.

4. PRODUCT OBJECTIVES

MECHSOURCE should:

1. Make finding spare parts easier.
2. Reduce incorrect part purchases.
3. Improve access to verified suppliers.
4. Make supplier and price comparison easier.
5. Connect customers with mechanics and technicians.
6. Improve access to workshops.
7. Support vehicle and equipment maintenance.
8. Support fleet operators.
9. Support industrial procurement.
10. Create a scalable marketplace for machinery related products and services.
11. Use structured data and AI to improve part identification and discovery.
12. Create a platform capable of expanding across Nigeria, Africa and eventually international markets.

5. TARGET USERS, MECHSOURCE will serve multiple user groups.

5.1 Individual Vehicle Owners: People who own cars, trucks, motorcycles or other vehicles and need parts, maintenance or repair services.

5.2 Fleet Operators: Businesses managing multiple vehicles that need parts, maintenance, servicing, repair and procurement.

5.3 Truck Owners and Operators: Users requiring commercial vehicle parts, diesel engine parts, tyres, batteries, filters, lubricants, repairs and specialist services.

5.4 Heavy Equipment Owners: Businesses and individuals operating construction, mining, agricultural or industrial machinery.

5.5 Mechanics: Independent mechanics looking for customers, parts and specialist services.

5.6 Specialist Technicians: Professionals specialising in diesel engines, electrical systems, hydraulics, heavy equipment, diagnostics and other technical areas.

5.7 Workshops: Automotive and machinery repair businesses requiring customers, parts and business management tools.

5.8 Spare Parts Suppliers: Businesses selling automotive, truck, diesel engine, heavy equipment and industrial parts.

5.9 Equipment Suppliers: Businesses selling construction, mining, agricultural and industrial equipment.

5.10 Equipment Rental Businesses: Companies renting machinery and equipment.

5.11 Industrial Businesses: Businesses requiring machinery parts, tools, PPE, lubricants, electrical components and industrial equipment.

5.12 Procurement Teams: Organisations purchasing large quantities of parts, tools, machinery or maintenance supplies.

5.13 Logistics Providers: Businesses responsible for transporting purchased products and equipment.

5.14 Platform Administrators: MECHSOURCE staff responsible for operating and managing the platform.

6. USER ROLES

The platform should support role based experiences.

Primary roles include:

- Customer
- Supplier
- Technician
- Workshop
- Equipment Provider
- Rental Provider
- Fleet Manager
- Procurement User
- Logistics Provider
- Administrator

A user may eventually have more than one business relationship with the platform.

The architecture should therefore avoid unnecessarily restricting users to a single permanent role (except admin).

7. CORE PLATFORM AREAS

MECHSOURCE will consist of several connected platform areas.

1. Parts Marketplace
2. Equipment Marketplace
3. Tools Marketplace
4. PPE and Safety Marketplace
5. Lubricants and Filters Marketplace
6. Batteries and Electrical Marketplace
7. Technician Marketplace
8. Workshop Directory
9. Equipment Rental
10. Emergency Breakdown Services
11. Find My Part
12. AI Parts Identification
13. Maintenance Management
14. Fleet Management
15. B2B Procurement
16. Supplier Management
17. Inventory Management
18. Orders and Delivery
19. Payments
20. Reviews and Ratings
21. Search
22. Notifications
23. Customer Support
24. Administration
25. Analytics

26. PARTS MARKETPLACE

The Parts Marketplace is one of the core components of MECHSOURCE.

It should support categories including:

- Car Parts
- Truck Parts
- Diesel Engine Parts
- Heavy Equipment Parts
- Construction Equipment Parts
- Mining Equipment Parts
- Agricultural Equipment Parts
- Industrial Equipment Parts
- Filters
- Lubricants
- Batteries
- Electrical Components
- Workshop Tools
- PPE and Safety Equipment

The marketplace should support:

- Product discovery
- Search
- Filtering
- Sorting
- Compatibility

- Supplier comparison
- Price comparison
- Stock information
- Supplier information
- Product reviews
- Delivery information
- Cart
- Checkout
- Orders
- Returns where applicable

9. PRODUCT INFORMATION

A product listing should be capable of displaying:

- Product name
- Product images
- Description
- Manufacturer
- OEM number
- Manufacturer part number
- Alternative part numbers
- Compatible vehicles
- Compatible equipment
- Compatible engines
- Price
- Stock status
- Quantity
- Supplier
- Supplier location
- Supplier rating
- Warranty information
- Return information
- Delivery options
- Estimated delivery time

The interface should clearly distinguish verified compatibility from supplier provided information where necessary.

10. FIND MY PART

Find My Part should be one of the most important MECHSOURCE experiences.
The system should support several identification methods.

Method 1: Vehicle or Equipment Information

User selects:

- Manufacturer
- Model
- Year
- Engine
- Transmission where relevant
- Part category
- Part required

Method 2: OEM Number

User enters an OEM or manufacturer part number.

Method 3: Natural Language Search

User describes what they need in ordinary language.

Example:

I need a fuel filter for a 2018 Hilux 2.8 diesel near Ikeja.

Method 4: Image Identification

User uploads or photographs a part.

The system provides possible identification.

Method 5: Human Verification

User can request assistance from a verified supplier, technician or appropriate specialist where identification is uncertain.

11. AI PART IDENTIFICATION

AI should support the platform but should not replace professional verification where accuracy is critical.

Potential AI capabilities include:

- Image based part recognition
- Natural language part search
- Part classification
- Compatibility assistance
- OEM number matching
- Alternative part suggestions
- Supplier matching
- Product recommendations
- Maintenance assistance
- Procurement assistance

AI generated results should include appropriate confidence or verification indicators where applicable.

12. COMPATIBILITY SYSTEM

MECHSOURCE must be designed around compatibility.

The platform should eventually be able to establish relationships between:

- Manufacturer
- Model
- Year
- Generation
- Engine
- Engine code
- Transmission
- Vehicle
- Equipment
- Part
- OEM number
- Alternative part
- Supplier

The system should not assume that two parts are compatible merely because their names are similar.

Compatibility information should be based on verified data wherever possible.

13. SUPPLIER MARKETPLACE

Suppliers should have dedicated business functionality.

Supplier capabilities should include:

- Business registration
- Business verification
- Business profile
- Business location
- Product listing
- Inventory management
- Pricing
- Product images
- OEM information
- Compatibility information
- Order management
- Delivery management
- Customer communication
- Reviews
- Sales history
- Analytics

14. SUPPLIER VERIFICATION

The platform should support supplier verification.

Potential verification information may include:

- Business identity
- Business registration information
- Contact information
- Physical business location
- Business category
- Supporting documents
- Product authenticity information where appropriate

The exact verification process will be defined during the research and business requirements stage.

15. TECHNICIAN MARKETPLACE

Customers should eventually be able to find qualified technicians.

Technician profiles should include:

- Name
- Professional category
- Specialisation
- Experience
- Location
- Services
- Availability
- Service area
- Pricing where applicable
- Verification status
- Ratings

Reviews
Completed jobs

Examples of specialist categories include:

General mechanic
Diesel mechanic
Auto electrician
Heavy equipment technician
Hydraulic technician
Engine specialist
Transmission specialist
Diagnostic technician

16. WORKSHOP DIRECTORY

Customers should be able to find workshops based on:

Location
Specialisation
Vehicle type
Equipment type
Services
Availability
Rating
Verification status

Workshop profiles should include:

Workshop name
Location
Services
Specialisations
Operating information
Technicians
Contact methods
Ratings
Reviews
Verification

17. EMERGENCY BREAKDOWN SERVICES

MECHSOURCE should eventually support emergency assistance.

A customer with a breakdown could:

Request assistance
Share location
Describe the problem
Upload a photograph
Select vehicle or equipment
View nearby available technicians
Request assistance
Track the service request

This feature should be designed for future integration without requiring the core marketplace to be rebuilt.

18. EQUIPMENT MARKETPLACE

MECHSOURCE should eventually support equipment buying and selling.

Categories may include:

- Excavators
- Loaders
- Bulldozers
- Cranes
- Generators
- Compressors
- Agricultural machinery
- Mining machinery
- Construction machinery
- Industrial machinery
- Heavy trucks

Equipment listings should support:

- Equipment details
- Manufacturer
- Model
- Year
- Condition
- Usage information where relevant
- Location
- Price
- Images
- Seller information
- Inspection information
- Contact or enquiry

19. EQUIPMENT RENTAL

The platform should eventually support equipment rental.

Users should be able to:

- Search equipment
- Filter by location
- View equipment
- View rental terms
- View availability
- Request rental
- Make payment where applicable
- Track rental status
- Return equipment

Rental providers should have inventory and booking management.

20. MAINTENANCE MANAGEMENT

MECHSOURCE should eventually allow users and businesses to maintain digital records for vehicles and equipment.

Maintenance records may include:

- Vehicle or equipment
- Service date
- Service type
- Parts replaced
- Technician
- Workshop
- Cost
- Mileage or operating hours
- Notes
- Documents
- Next service date

Users should receive maintenance reminders where applicable.

21. FLEET MANAGEMENT

Fleet management should allow businesses to manage multiple vehicles.

Potential capabilities include:

- Fleet dashboard
- Vehicle records
- Driver records
- Mileage
- Maintenance
- Repairs
- Parts
- Service schedules
- Maintenance costs
- Vehicle status
- Documents
- Reports
- Notifications

22. B2B PROCUREMENT

Businesses should eventually be able to create procurement requests.

Example:

A company requires 50 fuel filters for a specific fleet.

The procurement workflow could be:

- Create request
 - Specify requirements
 - Specify quantity
 - Submit request
 - Suppliers respond
 - Compare quotations
 - Select supplier
 - Place order
 - Payment

- Delivery
- Completion

The system should be capable of supporting both small and large procurement requirements.

23. SEARCH

Search is a major platform capability.

Users should be able to search using:

- Part name
- OEM number
- Manufacturer number
- Vehicle
- Equipment
- Supplier
- Technician
- Workshop
- Natural language

Search should eventually understand relationships between products, vehicles, equipment and suppliers.

24. LOCATION SERVICES

Location will be important throughout MECHSOURCE.

Location functionality may support:

- Supplier discovery
- Workshop discovery
- Technician discovery
- Equipment discovery
- Emergency services
- Delivery
- Equipment rental
- Search by distance
- Service area

Users should have control over location permissions and manually enter locations where necessary.

25. CART AND CHECKOUT

Customers should be able to:

- Add products
- Change quantities
- Remove products
- Review supplier information
- Review delivery options
- Review prices
- Apply applicable discounts
- Select delivery address
- Select payment method
- Confirm order
- Receive confirmation

The architecture should account for situations where products in one order come from different suppliers.

26. PAYMENTS

The platform should support secure payment processing.

Potential capabilities include:

- Customer payments
- Supplier settlement
- Marketplace commission
- Refunds
- Payment verification
- Transaction records
- B2B payments

The final payment providers will be determined during the business and technical research stages.

27. ORDERS

Customers should be able to view:

- Order number
- Products
- Supplier
- Quantity
- Price
- Payment status
- Order status
- Delivery status
- Tracking information
- Estimated delivery
- Order history

Suppliers should be able to manage orders from their own interface.

28. DELIVERY AND LOGISTICS

MECHSOURCE should support:

- Delivery addresses
- Supplier location
- Customer location
- Delivery options
- Delivery charges
- Delivery status
- Estimated delivery time
- Tracking
- Proof of delivery

The architecture should allow integration with multiple logistics providers.

29. REVIEWS AND RATINGS

The platform should support reviews for:

- Products
- Suppliers
- Technicians
- Workshops
- Equipment providers

Rental providers

Reviews should be associated with appropriate transactions or service interactions where possible to reduce fraudulent reviews.

30. NOTIFICATIONS

Notifications may include:

- Order confirmation
- Payment confirmation
- Order status
- Delivery status
- Price changes
- Stock availability
- Technician booking
- Maintenance reminders
- Procurement updates
- Messages
- Platform announcements

31. CUSTOMER SUPPORT

Customers should have access to support.

Potential support channels include:

- Help centre
- FAQs
- Order support
- Supplier disputes
- Payment support
- Delivery support
- Technician support
- Contact support

The final support architecture will be determined during later stages.

32. MESSAGING

MECHSOURCE may require controlled communication between:

- Customer and supplier
- Customer and technician
- Customer and workshop
- Business and supplier
- Procurement user and supplier

Messaging should be designed around privacy, security and transaction context.

33. ADMINISTRATION

MECHSOURCE requires a central administration platform.

Administrators should be able to manage:

- Users
- Suppliers
- Technicians

- Workshops
- Products
- Parts
- Vehicles
- Equipment
- Compatibility data
- Orders
- Payments
- Deliveries
- Reviews
- Disputes
- Verification
- AI identification reviews
- Categories
- Notifications
- Promotions
- Analytics
- Platform settings

34. ANALYTICS

The platform should eventually provide analytics such as:

- Most searched parts
- Most requested vehicles
- Most requested equipment
- Popular suppliers
- Demand by location
- Average prices
- Stock shortages
- Order volumes
- Supplier performance
- Delivery performance
- Customer activity
- Technician performance
- Marketplace revenue

35. TRUST AND SAFETY

MECHSOURCE must be designed around trust.

The platform should support:

- Supplier verification
- Technician verification
- Workshop verification
- Product information verification
- Transaction records
- Reviews
- Ratings
- Dispute handling
- Fraud monitoring
- Return policies
- Refund policies

The exact operational policies will be established during later research.

36. USER EXPERIENCE PRINCIPLES

MECHSOURCE should be:

- Simple
- Clear
- Fast
- Trustworthy
- Professional
- Accessible
- Mobile first
- Scalable
- Easy to navigate
- Responsive

The platform may contain complex technical information, but the customer interface should not feel complicated.

A normal vehicle owner should be able to use MECHSOURCE without being an automotive expert.

37. MOBILE APPLICATION

The primary customer product will be a mobile application supporting:

- Android
- iOS

The application should eventually provide access to the customer marketplace and relevant services.

A cross platform approach may be considered during technical architecture planning.

The mobile experience should be designed mobile first.

38. WEB PLATFORM

MECHSOURCE will require web based experiences for appropriate users.

These may include:

- Supplier dashboard
- Technician dashboard
- Workshop dashboard
- Business dashboard
- Fleet management
- Procurement
- Administration

The final separation between mobile and web experiences will be established during UX and technical architecture planning.

39. PLATFORM ARCHITECTURE PRINCIPLE

MECHSOURCE must be treated as a platform rather than a single application.

The customer mobile application is only one interface into the MECHSOURCE ecosystem.

The platform should eventually consist of:

- Customer Platform
- Supplier Platform
- Technician Platform

Workshop Platform
Equipment Platform
Fleet Platform
Maintenance Platform
Procurement Platform
Administration Platform
Parts and Compatibility Data Platform
AI Platform
Marketplace Platform
Logistics Integration
Payment Infrastructure

40. SCALABILITY REQUIREMENT

The platform should be capable of expanding from:

Lagos

to

Ogun

Abuja

Other Nigerian states

Nigeria

West Africa

Africa

International markets

The architecture must therefore avoid assumptions that permanently restrict the platform to one city or country.

41. INTERNATIONAL EXPANSION

Future expansion may require support for:

- Multiple countries
- Multiple currencies
- Multiple languages
- Different tax systems
- Different payment providers
- Different logistics providers
- Different regulatory requirements
- Different supplier structures

These requirements should be considered during architecture planning.

42. DATA DEPENDENCY

MECHSOURCE is highly dependent on structured and reliable data.

The platform may require data relating to:

- Vehicles
- Equipment
- Parts
- OEM numbers
- Manufacturer numbers
- Compatibility
- Suppliers
- Inventory

- Prices
- Locations
- Technicians
- Workshops

The source, licensing, collection, validation and maintenance of this data will be formally investigated in Document 02.

43. AI DEPENDENCY

AI is an important component of MECHSOURCE but should not be the sole foundation of the platform.

The core platform must remain useful even when AI cannot confidently identify a part.

AI should enhance:

- Discovery
- Identification
- Search
- Recommendations
- Matching
- Maintenance assistance
- Procurement

AI should not be presented as infallible.

44. PRODUCT TRUST MODEL

The platform should distinguish between:

- Verified information
- Supplier provided information
- AI generated suggestions
- Customer provided information
- Technician verified information

The interface should make these distinctions clear where they affect purchasing decisions.

45. BUSINESS MODEL

Potential revenue models include:

- Marketplace commission
- Supplier subscriptions
- Featured listings
- Transaction fees
- Technician service commission
- Workshop services
- Equipment rental commission
- B2B procurement fees
- Fleet management subscriptions
- Maintenance management subscriptions
- Premium business tools

The final commercial model will be established through market research and business validation.

46. PHASED PRODUCT DEVELOPMENT

MECHSOURCE should be developed progressively.

Phase 1:

Parts marketplace
Find My Part
Customer accounts
Vehicle profiles
Supplier marketplace
Search
Product comparison
Orders
Payments
Delivery
Reviews
Basic technician discovery

Phase 2:

Heavy equipment
Specialist technicians
Workshop services
Emergency breakdown services
Equipment marketplace
Equipment rental

Phase 3:

Fleet management
Maintenance management
B2B procurement
Advanced supplier tools
Advanced analytics

Phase 4:

International expansion
Advanced AI
Additional industrial categories
Additional business services

47. UI/UX REQUIREMENT

UI/UX must be designed before software development.

The design process must include:

- Information architecture
- User journeys
- User flows
- Screen inventory
- Wireframes
- Design system
- High fidelity designs
- Responsive layouts
- Interactive prototype
- UX validation

The Figma design should represent the intended product accurately enough to serve as a reference for future development.

48. TECHNICAL DEVELOPMENT REQUIREMENT

No software development should begin until the following have been completed and reviewed:

- Product requirements
- Research and data requirements
- UX/UI specification
- Figma design
- Technical architecture

The final development process must follow the approved product and design specifications.

49. SUCCESS CRITERIA

MECHSOURCE should ultimately succeed when customers can reliably:

- Identify what part they need
- Determine whether it is compatible
- Find trustworthy suppliers
- Compare available options
- Purchase the correct part
- Receive the part conveniently
- Find appropriate technical assistance
- Maintain their vehicles or equipment digitally

Businesses should be able to:

- Manage suppliers
- Manage inventory
- Receive orders
- Find customers
- Manage technicians
- Manage workshops
- Manage fleets
- Manage maintenance
- Submit procurement requests
- Track business activity

50. LONG TERM VISION

The long term objective is to make MECHSOURCE a major digital infrastructure layer for vehicle, machinery and equipment ownership and maintenance.

The platform should move beyond simply selling parts.

The long term ecosystem is:

- Identify
- Find
- Compare
- Buy
- Deliver

- Repair
- Maintain
- Manage
- Rent

- Procure
- Operate

The platform should become a trusted digital connection between the people, businesses, products, services and data required to keep vehicles, machinery and equipment operational.

51. PRODUCT DEFINITION

MECHSOURCE is therefore defined as:

A scalable digital marketplace and maintenance infrastructure platform connecting vehicle owners, equipment owners, businesses, suppliers, technicians, workshops and industrial service providers through intelligent part identification, marketplace discovery, procurement, maintenance and service management.

The mobile application is the primary customer interface.

The wider MECHSOURCE platform is the underlying ecosystem.

This definition should remain the central reference point for subsequent research, data planning, UX/UI design, technical architecture and development.